

Shahnewaz Leon

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EDUCATION

Ph.D. in Computer Science (CGPA: 4.0/4.0)

Expected Dec 2026

North Carolina State University (NCSU), Raleigh, NC, USA

Research Focus: Sequential Recommendation, Graph-Augmented Search & Retrieval, Predictive Behavioral Modeling

M.Sc. in Computer Science (CGPA: 4.0/4.0)

Dec 2024

North Carolina State University (NCSU), Raleigh, NC, USA

SKILLS

Programming Languages: Python, C++, Java, SQL, Go, JavaScript, NodeJS, Linux Shell Scripting

Machine Learning & AI: PyTorch, TensorFlow, Transformers, Scikit-Learn, Pandas, NumPy

Distributed Systems & Backend: Microservices, Event-Driven Architecture, AWS, Docker, Kubernetes, Terraform

Databases & Storage Engines: MySQL, Postgres, Elasticsearch, Redis, MongoDB, DynamoDB, Oracle

Engineering Practices: CI/CD (GitHub Actions, Jenkins, AWS CodeBuild), Test Automation (PyTest, JUnit, Mockito), Code Review & Mentorship

APPLIED ML RESEARCH & SYSTEMS EXPERIENCE

Graduate Research Assistant (NC State University, USA)

Aug 2022 – Present

- **Collective Priors (VL/HCC '26):** Designed and evaluated a sequential recommendation architecture utilizing Markov state transitions and graph reachability; achieved 0.804 Hit@10, outperforming frontier LLM baselines by over 16 percentage points in cold-start prediction.
- **PFIS-T (CHI '26 - Honorable Mention):** Engineered a predictive sequence recommendation model integrating temporal user interaction signals and social network topologies; achieved 81.5% next-step prediction accuracy, outperforming baseline models by 16.7%.
- **Learning to Navigate (HCII '26):** Engineered an end-to-end ML pipeline for repository-aware code search utilizing learned beam search over structured graphs; achieved MRR 0.91 / R@1 0.86.

Instructor of Record – CSC 216: Software Dev Fundamentals (NC State University, USA)

May 2026 – Aug 2026

- Directed curriculum delivery and served as sole Instructor of Record for the foundational software engineering course, managing teaching assistants and student code evaluations.

PROFESSIONAL ENGINEERING EXPERIENCE

Senior Software Engineer (Monstarlab, Bangladesh)

Aug 2017 - Jul 2022

- **High-Concurrency Microservices:** Architected event-driven distributed microservices and automated CI/CD infrastructure using Terraform, GitHub Actions, and AWS.
- **Performance & Latency Optimization:** Owned end-to-end data aggregation pipelines for large-scale enterprise services using Spring Boot; configured proactive telemetry and CloudWatch alerting to resolve system bottlenecks, reducing production API latency by 50%.
- **Serverless Event Pipelines:** Engineered scalable recruitment pipelines utilizing AWS Lambda and EventBridge, minimizing operational costs and ensuring fault tolerance across distributed workflows.
- **Architecture & Leadership:** Defined High-Level and Low-Level Designs (HLD/LLD), led technical specification reviews, and mentored junior engineers on clean code and testing standards.

Backend Developer (Ekhanai.com, Bangladesh)

Apr 2016 - Jul 2017

- **Large-Scale Search & Retrieval:** Scaled distributed backend infrastructure and real-time search indexing for an e-commerce platform serving over 1 million daily active users (1M+ DAU), deploying Elasticsearch and MySQL to eliminate query latency bottlenecks.

Software Developer (SekaiLab Bangladesh, Bangladesh)

Jul 2015 - Mar 2016

- **Graph Visualization & Relational Data:** Engineered a dynamic influence mapping and graph analytics engine within an HR platform, processing relational data to identify high-impact organizational connectivity.

Software Engineer (Fronture Technologies, Bangladesh)

Feb 2014 - Nov 2014

- Automated CI build process workflows in Team Foundation Server and developed backend RESTful services for multi-platform enterprise time-tracking applications.